

15 C

$$E_T = \frac{1}{2} m \omega^2 x_o^2$$

At $t = 0$ s, $x_o = 1.50$ cm

$$E_T = \frac{1}{2} (0.5) (2\pi/0.5)^2 (1.50 \times 10^{-2})^2 = 8.88 \times 10^{-3} \text{ J}$$

At $t = 0$ s, $x_o = 0.65$ cm

$$E_T = \frac{1}{2} (0.5) (2\pi/0.5)^2 (0.65 \times 10^{-2})^2 = 1.67 \times 10^{-3} \text{ J}$$

$$\text{Decrease in energy} = 8.88 \times 10^{-3} - 1.67 \times 10^{-3} = 7.2 \times 10^{-3} \text{ J}$$